

Collaborator or Assistant?

How AI Coding Agents Partition Work Across Pull Request Lifecycles

Agent-associated corpus. Human-surfaced trace. Governance ambiguity made measurable.

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AI-associated code is entering popular repositories faster than accountability traces can explain it.

29,585
agent-associated
PRs
in 100+ star repos

NO RECORDED HUMAN-APPROVED REVIEW TRACE

89.6%

of terminal PRs lack a human-approved review.
Among merged PRs, only 12.8% show one.

TOOL-LEVEL REVIEW SPREAD

17x

Copilot 66.4% vs. OpenAI Codex 3.9%.

POINT 1

Agency visibility is the first governance problem

The paper's lifecycle map lets us separate what the repository records from what we might wish it recorded.

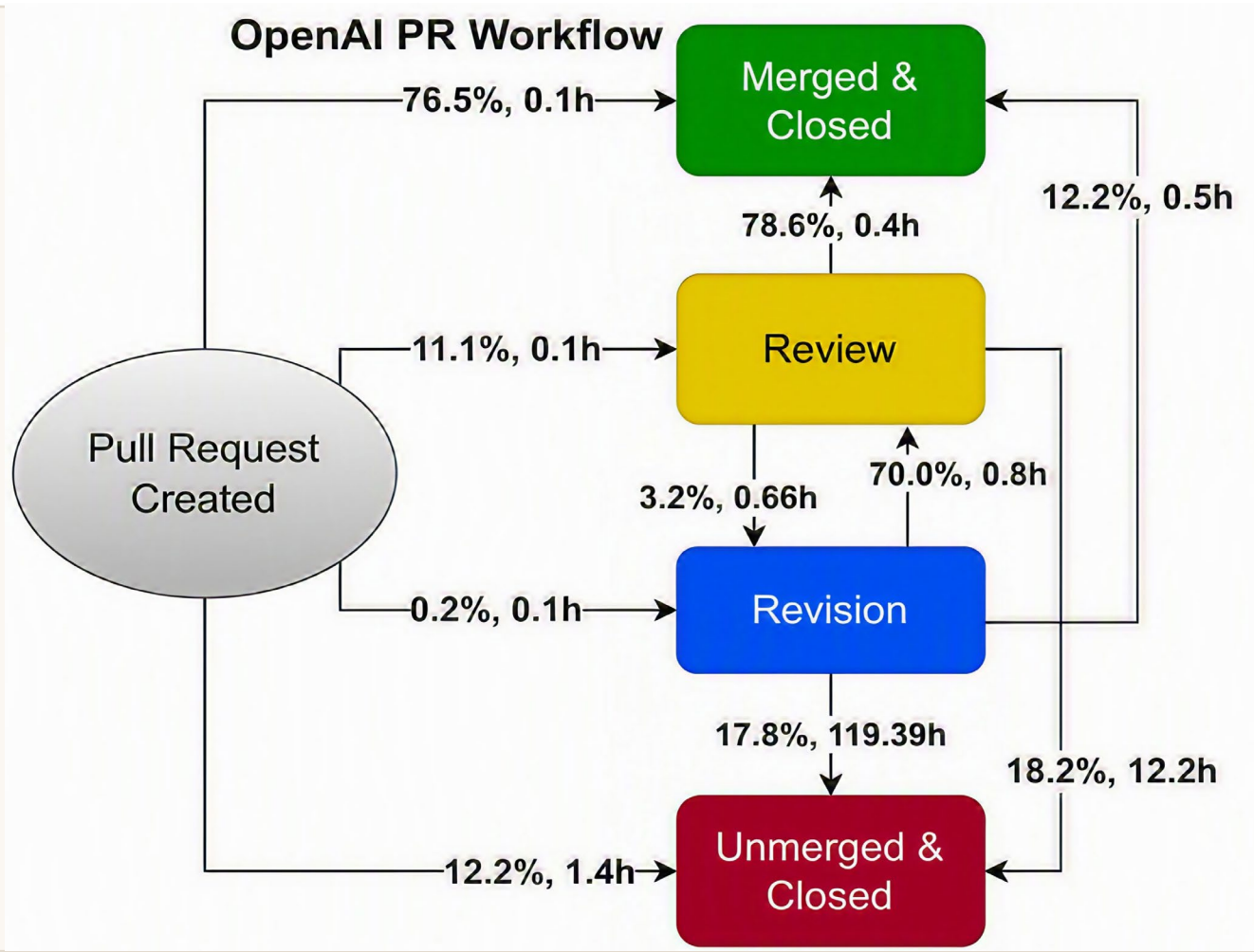
The trace records surfaces

GitHub logs accounts, commits, reviews, and merges. It does not reliably identify whether a human, an AI agent, or a human using an agent actually produced the code or made the decision.

Codex and Claude expose the gap

Li's attribution work shows that some tools leave visible bot signals, while Codex and Claude can surface through human-looking commit attribution. The log records the signer, not necessarily the initiator or accountable actor.

Workflow values show lifecycle paths, not approval rates.



POINT 2

Agency is real, but only partially legible

AIDev marks these PRs as AI-associated; AIWare reads what the platform trace makes visible. For Codex and Claude, Human-Init means human-surfaced first-commit attribution, not verified human authorship. The contribution is the separation: agent association, visible attribution, recorded review, merge execution.

Tool	Agent visibility in first-commit trace	Recorded human-approved review trace
Copilot	99.2%	66.4%
Devin	98.4%	44.7%
Cursor	96.1%	28.1%
Claude	AIDev 100% 4.1%*	19.4%
OpenAI Codex	AIDev 100% 0.1%	3.9%

*Claude is the fragile case: n=338 and 57% of labels flip under attribution-aware re-detection. Codex carries the obscured-agency pattern at scale (n=20,857).

Codex and Claude expose the trace problem: AIDev (Li et al., MSR 2026) accounts for these PRs as Agentic-PRs, yet the first-commit trace surfaces them as human-attributed in 99.9% and 95.9% of cases. Agent visibility appears in only 0.1% and 4.1%.

The empirical hinge

The paper matters because it gives us a way to read PR lifecycles as governance traces. Once we read the traces carefully, the weak spot is not simply productivity or authorship. It is accountability visibility.

POINT 3

The risk is not just code quality. It is accountable governance.

We do not prove these PRs are insecure. The alarm is that popular projects can merge AI-associated code while the trace may not show who produced it, whether accountable review occurred, or who answers if failure appears later.

Popular software surface

The analytic corpus is restricted to repositories with 100+ stars. These projects are more likely to be installed, reused, or depended on by downstream developers and users.

Thin recorded review

Across 29,585 terminal PRs, 89.6% have no recorded human-approved review trace. Among merged PRs, only 12.8% show one.

Accountability breaks

If a defect or security issue later appears, the repository trace may not establish whether the change was human-authored, agent-authored, or human-assisted agent work.

The missing link

trace → actor → accountable decision

The paper's significance: lifecycle traces reveal an accountability gap in AI-assisted software engineering.

AIDev: agent-associated PRs in popular repositories.

The finding: weak agency visibility aligns with thin recorded human-approved review.

The question: what should platforms/tools log before AI-assisted code enters?



Scan for code and data